

Indian semiconductor design market is expected to grow to US\$ 14.5 billion in 2015.

Over the years, the Indian semiconductor design industry has developed significant capabilities and currently works on state-of-the-art products. Growing technical competence, combined with stringent IP protection environment and trend towards localisation of contents, will help this segment continue to grow.

## Electronic manufacturing services

The growth of the electronic products market has led EMS companies to offer lower-cost manufacturing, faster turnaround for designs and quicker time to market. Since the mid-nineties, many global EMS companies have established operations in India and have become a key contributor in enhancing the eco-system in the country. Most of them are export-oriented and have encouraged local production using the same components and services base for the Indian market. Select electronics companies utilise the services of EMS companies to reduce their own investment and focus on their product development and marketing functions. These companies face competition from peers in other Asian countries such as China, Malaysia and Singapore.

The EMS industry services' contribution is expected to grow at a CAGR of 27.8 per cent from US\$ 0.4 billion to US\$ 1 billion in 2015. The overall turnover of the EMS industry is estimated to be US\$ 3 billion in 2011. Both the conventional and emerging industry segments are anticipated to maximise the benefits from this industry.

## Highlights of the national policy on electronics

India is one of the fastest-growing markets of electronics in the world. The demand for electronics in the Indian market was US\$ 45 billion in 2008-09 and is expected to reach US\$ 400 billion by 2020. The domestic production in 2008-09 was about US\$ 20 billion. It is expected that there will be a demand-

## MSME Performance in India

### Overall performance of the MSME sector

| Year     | Total MSMEs (million) | Market value of fixed assets (₹ billion) | Employment (million) | Gross output (₹ billion) |
|----------|-----------------------|------------------------------------------|----------------------|--------------------------|
| 2004-05  | 11.8                  | 1786.99                                  | 28.26                | 4297.96                  |
| 2005-06  | 12.34                 | 1881.13                                  | 29.49                | 4978.42                  |
| 2006-07  | 36.17                 | 8685.43*                                 | 80.52                | 13,513.83*               |
| 2007-08P | 37.73                 | 9174.37                                  | 84.23                | 14,351.79                |
| 2008-09P | 39.37                 | 9714.07                                  | 88.11                | 15,242.34                |
| 2009-10P | 41.08                 | 10,293.31                                | 92.22                | 16,193.55                |
| 2010-11P | 42.87                 | 10,948.93                                | 96.56                | 17,215.53                |
| 2011-12P | 44.78                 | 11,769.39                                | 101.26               | 18,343.32                |

Source: Ministry of MSME, Government of India

\*Estimated on the basis of per enterprise value obtained from sample survey of unregistered sector for activities in wholesale/retail trade, legal, education and social services, hotel and restaurants, transports and storage and warehousing (except cold storage) excluded from Fourth All India Census of MSME, unregistered sector. P=Projected

supply gap of nearly US\$ 300 billion by 2020. Indian electronics hardware production constitutes only around 1.31 per cent of the global production.

Although India is a recognised global player in the software and software services sector, it lags behind in electronics hardware manufacturing capabilities. India is increasingly becoming a destination for chip design and embedded software. The government's vision is to transform India into a global hub for ESDM to meet the growing domestic and global demand.

Given India's growing strength in chip design and embedded software, the increasing focus on design in product development has the potential to make India a favoured destination for ESDM. There is potential to use capabilities so created to successfully export ESDM products from India to other countries. The national policy on electronics aims to transform India into a premier ESDM hub.

The national policy on electronics seeks to:

1. Develop capacities for manufacture of strategic electronics in India.
2. Promote a vibrant and sustainable ecosystem of R&D, design and engineering and innovation to enhance manufacturing capabilities in electronic raw materials, components, sub assemblies as well as products.

3. Develop high-quality electronic products at affordable prices for adoption and deployment to improve productivity, efficiency and ease of operations in other sectors.

4. Provide attractive fiscal incentives across value chain of ESDM sector to eliminate disability costs in manufacturing and mitigate relatively high cost of finance.

5. Facilitate setting-up of semiconductor wafer fab facilities and its eco-system for design and fabrication of chips and chip components.

6. Provide incentives for setting up over 200 electronic manufacturing clusters (EMCs) with world-class logistics and infrastructure and easy-to-do business facilities.

7. Provide assistance to set-up 'Greenfield EMCs' and upgrade 'Brownfield EMCs.'

8. Support development of appropriate infrastructure to support EMCs, including development of logistics hub, port-to-factory linkages, roads and highways.

9. Rationalise and establish industry-friendly stable tax regime conducive to attract global investments and encourage electronics sector through appropriate fiscal incentives and taxation mechanisms.

10. Provide a ten-year stable tax regime for ESDM sector.

11. Technology transfer, R&D and IPR creation through a friendly tax regime.

12. Facilitate foreign direct investment (FDI) in ESDM sector.

13. Promote manufacturing capacities for sourcing ESDM in strategic and core infrastructure sectors—defence, atomic energy, space, railways, power, telecommunications, etc.

14. Enable defence offset obligations for electronic procurements to meet